MYP MATHEMATICS · YEAR 1

Mathematics for the MYP

Powers, expanded form and order of operations · Chapter 2 revision proof

2

SECOND EDITION · STUDENT COURSEBOOK · MYP 1 ONLY

02

Powers, expanded form and order of operations

A power is a compressed instruction: it records how many copies of a factor to multiply. A shared order of operations keeps that compression unambiguous.

ENGAGE

A storage label says “10 TB”. A computer reports the same space as “about 9.09 TiB”. Has storage disappeared? And when someone writes $2 + 3 \times 4$, what must we agree before the expression has one meaning?

Framing the inquiry

Key concept: Form
Related concepts: Simplification, Quantity
Global context: Scientific and technical innovation

Statement of inquiry: Compressing repeated multiplication into a power, and agreeing on the order in which operations apply, allows complex calculations to be recorded compactly and reproduced reliably.

Questions to guide us

- **Factual:** What does 5^3 mean? Which operation comes first in $6 + 2 \times 3$?
- **Conceptual:** Why does an exponent of 1 leave the base unchanged? How can 2^{10} and 10^3 be compared fairly?
- **Debatable:** Should a calculator decide the meaning of an expression, or should the writer make the meaning visible?

ATL · Transfer. In order for a student to transfer a rule to a new situation, students must check whether the structure is the same, apply the rule carefully, and test the result with an example or counterexample.

GETTING READY

Learning intentions and prerequisite check

You are learning to

- Understand a power as repeated multiplication and identify its base and exponent.
- Use powers of ten in expanded form and return to standard form.
- Recognise squares and cubes, and find square roots and cube roots.
- Apply the order of operations, including brackets, indices and roots.
- Substitute values into expressions and formulas containing powers.
- Compare quantities written as powers and justify the comparison.

Success criteria

I can expand and evaluate a power without reading the exponent as a multiplier.

I can explain why $a^1 = a$ and, for $a \neq 0$, $a^0 = 1$.

I can move between standard form and expanded form with powers of ten.

I can reverse squaring and cubing with roots.

I can show one operation per line and respect brackets.

I can explain how brackets change the base of a power.

READY CHECK

Try these without a calculator. They check multiplication, place value and brackets.

R1 Calculate 6 × 7 and 9 × 9. READY

R2 Write 4,000 + 500 + 30 + 6 as a single number. READY

R3 Calculate 12 + 3 × 4. READY

R4 Write 3 × 3 × 3 as a product and evaluate it. READY

R5 What is the area of a square with side 8 cm? READY

R6 Insert brackets so that 2 + 4 × 3 has the value 18. READY

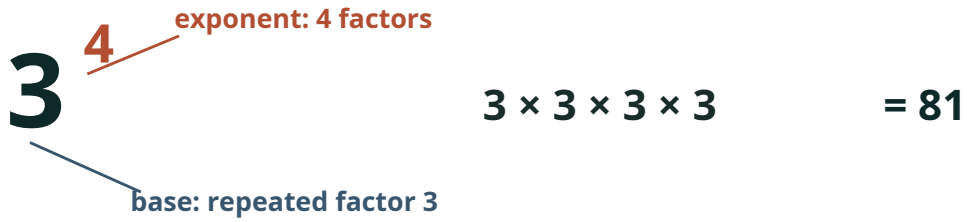
HOW TO USE THIS CHAPTER

Try the guided practice immediately after the matching example. Use the diagrams to explain what the notation means. Answers and student-facing reasoning are at the back of this proof. If more than two ready-check items are incorrect, review multiplication facts and brackets before continuing.

2.1 · FROM REPEATED MULTIPLICATION TO POWERS

The exponent counts factors

Writing $3 \times 3 \times 3 \times 3$ is correct but long. Index notation records two things: the repeated factor, called the **base**, and the number of copies, called the **exponent** or index.



The exponent counts factors; it is not a multiplier.

In 3^4, the base 3 is repeated four times. The value 81 is the result, not part of the notation.

WORKED EXAMPLE 1

Write each power as repeated multiplication and evaluate it: 2^5, 5^2, 10^4, 4^3.

1. $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.
2. $5^2 = 5 \times 5 = 25$. This is a square.
3. $10^4 = 10 \times 10 \times 10 \times 10 = 10{,}000$.
4. $4^3 = 4 \times 4 \times 4 = 64$. This is a cube.

Check: the number of factors must equal the exponent.

GUIDED PRACTICE · COPY AND COMPLETE

Use these immediately after the matching worked examples. Check your reasoning, not only the final number.

G1 In 8^3, identify the base and exponent, write the expanded product, and evaluate it. What would the value be if the exponent were 1? GUIDED

PRACTICE 2.1 · POWERS AND INDEX NOTATION

Show the expanded product at least once for each new power. Name the base and exponent before evaluating.

2.1.1 Write each product as a power. (a) $4 \times 4 \times 4$ (b) 7×7 (c) $10 \times 10 \times 10 \times 10 \times 10$ (d) $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$ FLUENCY

2.1.2 Evaluate. (a) 2^3 (b) 3^2 (c) 5^3 (d) 10^3 (e) 2^6 (f) 4^3 FLUENCY

2.1.3 Find the missing exponent. (a) $3^{\square} = 81$ (b) $2^{\square} = 64$ (c) $10^{\square} = 1{,}000$ (d) $7^{\square} = 1$ FLUENCY

2.1 · FROM REPEATED MULTIPLICATION TO POWERS

Special cases and common errors

An exponent of 1 means one factor: $7^1 = 7$. For a nonzero base, an exponent of 0 is defined to be 1. The pattern of dividing by the base explains the convention:

$$5^3 = 125, \quad 5^2 = 25, \quad 5^1 = 5, \quad 5^0 = 1.$$

MISCONCEPTION CHECK

$3^4 \neq 3 \times 4$. The exponent tells how many times to use the base as a factor. Likewise, $5^2 = 25$, not 10.

WORKED EXAMPLE 2

Write each product as a power and evaluate it: $6 \times 6 \times 6 \times 6$, 9×9, $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$.

1. Four factors of 6: $6^4 = 1{,}296$.
2. Two factors of 9: $9^2 = 81$.
3. Seven factors of 2: $2^7 = 128$.

Count the factors before writing the exponent.

PRACTICE 2.1 · POWERS AND INDEX NOTATION

Show the expanded product at least once for each new power. Name the base and exponent before evaluating.

2.1.4 Explain the difference between 4^2 and 4×2. Include the value of each. REASONING

2.1.5 A square tile has side length 9 cm. Write its area using a power and evaluate it. State the unit. APPLICATION

2.1.6 Compare 2^4 and 4^2. Are they equal? Compare 2^3 and 3^2. What does this suggest about the claim ‘swapping base and exponent never changes the value’? REASONING

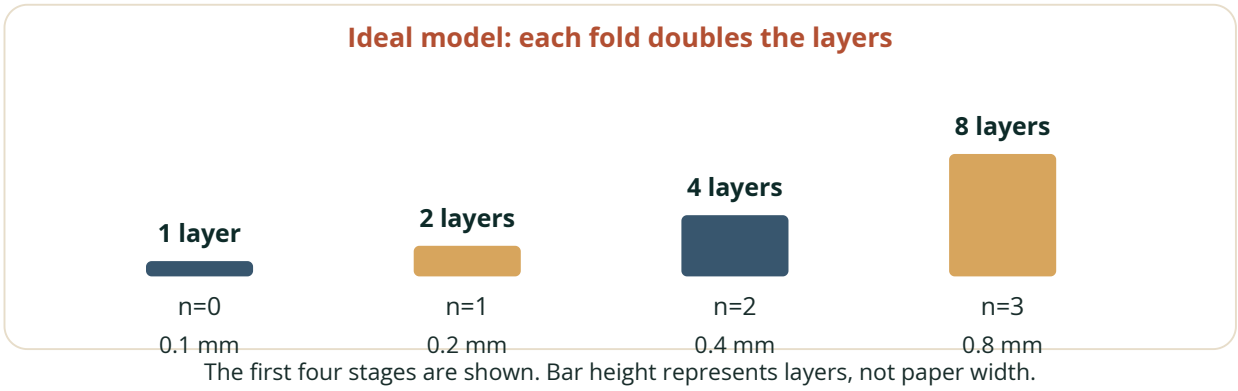
2.1.7 A rumour starts with one person. In round 1, that person tells 3 new people. In round 2, each of those 3 tells 3 new people, and the pattern continues. Write the number of new people told in round 4 as a power, evaluate it, and find the total number of new people told across rounds 1 to 4. APPLICATION

2.1.8 Decide which is larger, 2^{10} or 10^3, first by reasoning with factors and then by checking the values. CHALLENGE

EXPLORATION · CRITERION B

The doubling sheet

A sheet of paper is about 0.1 mm thick. In an **ideal model**, each fold doubles the number of layers and the thickness. Real paper may not allow the predicted number of folds.



The first four stages of the ideal model. Continue the pattern in the table.

FOLDS, N	LAYERS	THICKNESS / MM
0	1	0.1
1	2	0.2
2	4	
3		
4		
5		

EXPLORATION · THE DOUBLING SHEET

Criterion B practice. Separate what an ideal model predicts from what a physical sheet of paper allows. Use a table, state a rule, verify it, then test the model's limits.

B1 Fold a sheet of paper as many times as you safely can and record the number of folds you achieve. Then, under the ideal assumptions that every fold doubles the layers and the sheet is 0.1 mm thick, complete the layers and thickness for n = 0 to 5. INVESTIGATION

B2 Describe the pattern in words and write a rule for the thickness T millimetres after n folds. INVESTIGATION

B3 Verify your rule for n = 10. How many times thicker is the stack than before any folds? INVESTIGATION

EXPLORATION · CRITERION B

Test the model's limits

A rule can be correct inside an ideal model and still stop describing a physical object. Use your rule to make predictions, then state what the model assumes.

EXPLORATION · THE DOUBLING SHEET

Criterion B practice. Separate what an ideal model predicts from what a physical sheet of paper allows. Use a table, state a rule, verify it, then test the model's limits.

B4 Find the least number of folds for which the ideal stack is thicker than 1 m. Show the values on either side of your answer. INVESTIGATION

B5 People often say that a sheet of paper cannot be folded more than about eight times. Does the table prove that claim false? State two assumptions of the model and one way to improve the thickness estimate. REASONING

B6 Extension: a different ideal model triples in thickness with each fold, so $T = 0.1 \times 3^n$. Find the thickness after 5 folds and compare it with the doubling model. Which model grows faster, and why? CHALLENGE

ATL · Critical thinking. In order for a student to evaluate a model, students must distinguish the assumptions that make the rule work from the conditions needed for the rule to describe reality.

2.2 · POWERS OF TEN AND EXPANDED FORM

Place value is built from powers of ten

The ones place is 10^0, the tens place is 10^1, the hundreds place is 10^2, and so on. Each step left multiplies by 10; each step right divides by 10.

Millions	Hundred thousand	Ten thousands	Thousands	Hundreds	Tens	Ones
10^6	10^5	10^4	10^3	10^2	10^1	10^0
1,000,000	100,000	10,000	1,000	100	10	1

Each step left multiplies by 10; each step right divides by 10.

Place values from millions to ones written as powers of ten.

WORKED EXAMPLE 3

Write 5,207 in expanded form using powers of ten.

1. Read the places: 5 thousands, 2 hundreds, 0 tens, 7 ones.
2. Replace each place by a power of ten: $5 \times 10^3 + 2 \times 10^2 + 0 \times 10^1 + 7 \times 10^0$.
3. Check: $5{,}000 + 200 + 0 + 7 = 5{,}207$.

The zero term contributes nothing but preserves the place structure.

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

Keep zero placeholders visible when they help explain the place structure.

2.2.1 Evaluate. (a) 10^2 (b) 10^4 (c) 10^6 (d) 10^0 FLUENCY

2.2.2 Write 8,304 in expanded form using powers of ten. FLUENCY

2.2.3 Write $4 \times 10^3 + 7 \times 10^2 + 0 \times 10^1 + 6 \times 10^0$ in standard form. FLUENCY

2.2 · POWERS OF TEN AND EXPANDED FORM

Return to standard form without losing zeros

Expanded form is longer than standard form because it exposes the value of every digit. That makes it useful for checking place value and for seeing the effect of multiplying or dividing by 10.

WORKED EXAMPLE 4

Write $6 \times 10^4 + 0 \times 10^3 + 3 \times 10^2 + 9 \times 10^1 + 0 \times 10^0$ in standard form. State the value of the digit 3.

1. Evaluate nonzero terms: 60,000, 300, 90.
2. Keep the empty thousands and ones places: 60,390.
3. The 3 is in the hundreds place, so it is worth 300.

60,390; the digit 3 contributes 300.

MISCONCEPTION CHECK

10^4 is not 40. It is $10 \times 10 \times 10 \times 10 = 10{,}000$.

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

Keep zero placeholders visible when they help explain the place structure.

2.2.4 Write 90,050 in expanded form using powers of ten, then state the value of the digit 5. FLUENCY

2.2.5 A number is written as $2 \times 10^4 + 5 \times 10^1$. Explain why the standard form contains zeros, and write the number. REASONING

2.2.6 A simulated school roll is 12,480 students. Write this number in expanded form using powers of ten and explain which digit contributes the most to the total. APPLICATION

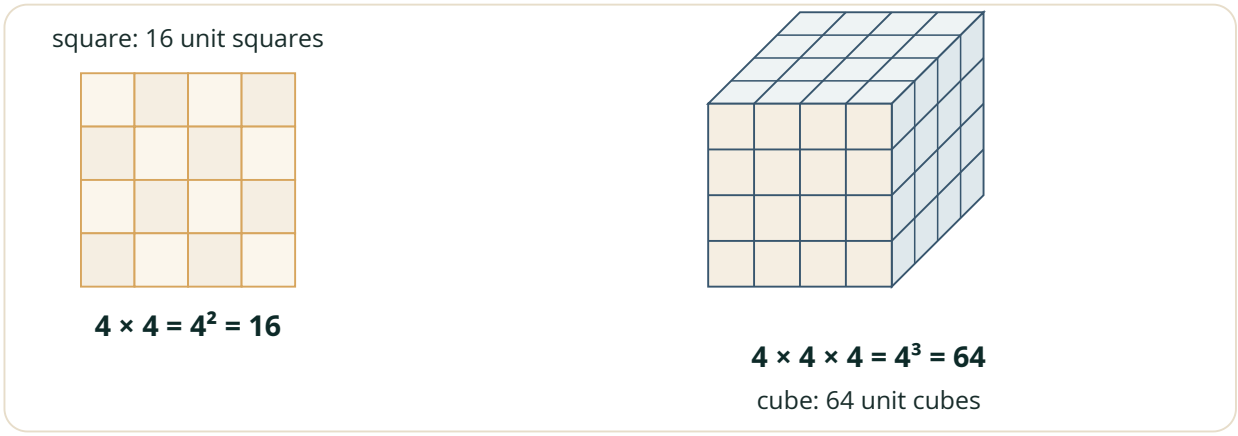
2.2.7 Compare 3×10^3 and 3×10^2. How many times larger is the first expression? Explain using place value. REASONING

2.2.8 Write a five-digit number whose expanded form contains exactly one zero term and whose digits add to 18. Explain how you constructed it and whether more than one answer is possible. CHALLENGE

2.3 · SQUARES, CUBES AND ROOTS

Models for squares and cubes

A **square number** is a whole number multiplied by itself. The positive square numbers begin 1, 4, 9, 16, … A **cube number** is a whole number multiplied by itself twice: 1, 8, 27, 64, …



A square shows two equal dimensions; a cube shows three. The cube can be seen as 4 layers of 16 unit cubes.

WORKED EXAMPLE 5

Evaluate 12^2, 7^3, $\sqrt{81}$, $\sqrt{144}$ and $\sqrt[3]{64}$.

1. $12^2 = 144$; $7^3 = 343$.
2. $9^2 = 81$, so $\sqrt{81} = 9$.
3. $12^2 = 144$, so $\sqrt{144} = 12$.
4. $4^3 = 64$, so $\sqrt[3]{64} = 4$.

Reverse a root by squaring or cubing the result.

GUIDED PRACTICE · COPY AND COMPLETE

Use these immediately after the matching worked examples. Check your reasoning, not only the final number.

G2 Since $13^2 = 169$, what is $\sqrt{169}$? Since $6^3 = 216$, what is $\sqrt[3]{216}$? A square has area 196 cm²; find its side length. GUIDED

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS

Check roots by reversing the operation: square a square root, or cube a cube root.

2.3.1 List the first twelve square numbers. FLUENCY

2.3.2 Evaluate. (a) 13^2 (b) 15^2 (c) 6^3 (d) 10^3 FLUENCY

2.3.3 Evaluate. (a) $\sqrt{49}$ (b) $\sqrt{100}$ (c) $\sqrt{169}$ (d) $\sqrt[3]{8}$ (e) $\sqrt[3]{27}$ (f) $\sqrt[3]{125}$ FLUENCY

2.3 · SQUARES, CUBES AND ROOTS

Roots reverse powers; they do not split sums

The square root symbol asks for the nonnegative value whose square is the number under the root. The cube root asks for the value whose cube is the number. In geometry, roots recover a side or edge from an area or volume.

WORKED EXAMPLE 6

A square has area 121 cm^2. A cube has volume 125 cm^3. Find the side or edge. If the cube's edge doubles, what happens to its volume?

1. $\sqrt{121} = 11$, so the square's side is 11 cm.
2. $\sqrt[3]{125} = 5$, so the cube's edge is 5 cm.
3. Doubling all three dimensions multiplies volume by $2^3 = 8$.

The new cube volume is 8 times the original.

MISCONCEPTION CHECK

$\sqrt{36 + 64} = \sqrt{100} = 10$, but $\sqrt{36} + \sqrt{64} = 6 + 8 = 14$. A root applies to the whole value underneath it.

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS

Check roots by reversing the operation: square a square root, or cube a cube root.

2.3.4 A square mosaic uses 144 identical tiles. How many tiles are along each side? APPLICATION

2.3.5 Find the positive value of x. (a) $x^2 = 64$ (b) $x^3 = 216$ (c) $x^2 = 225$ FLUENCY

2.3.6 Compare √81 + √16 with √(81 + 16). Are they equal? Explain the difference. REASONING

2.3.7 Between which two consecutive whole numbers does √50 lie? Explain using square numbers, then state which whole number it is closer to. CHALLENGE

2.3.8 A cube has volume 343 cm^3. Find its edge length. If the edge length is doubled, calculate the new volume and explain how your answer connects to 2^3. CHALLENGE

2.4 · ORDER OF OPERATIONS

A shared convention makes expressions readable

Without an agreed order, $2 + 3 \times 4$ could mean 20 or 14. Mathematics uses a convention so that the same expression has the same value for every reader.

PRIORITY	OPERATIONS	HOW TO APPLY IT
1	Brackets and grouping symbols	Work from the innermost bracket outwards. A fraction bar groups the numerator and denominator.
2	Indices and roots	Evaluate powers and roots before multiplication, division, addition or subtraction.
3	Multiplication and division	Same priority: work left to right.
4	Addition and subtraction	Same priority: work left to right.

WORKED EXAMPLE 7

Evaluate $7 + 2 \times 3^2$, $(7 + 2) \times 3^2$, $24 \div 4 \times 2$ and $24 \div (4 \times 2)$.

1. $7 + 2 \times 9 = 7 + 18 = 25$.
2. $9 \times 9 = 81$.
3. Left to right: $24 \div 4 = 6$, then $6 \times 2 = 12$.
4. Brackets first: $24 \div 8 = 3$.

25, 81, 12, 3.

PRACTICE 2.4 · ORDER OF OPERATIONS

Show one operation per line when an expression has several steps. Multiplication and division share a rank; addition and subtraction share a rank.

2.4.1 Evaluate $8 + 2 \times 5$. FLUENCY

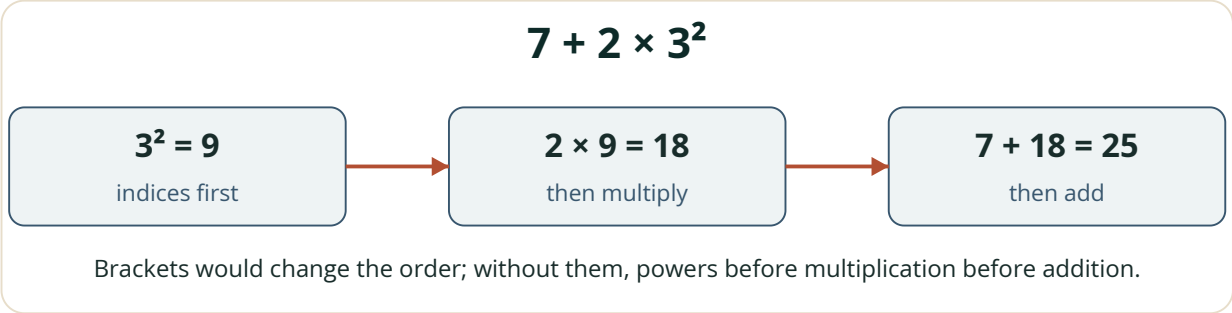
2.4.2 Evaluate $20 - 3 \times 4$. FLUENCY

2.4.3 Evaluate. (a) $6 + 4^2$ (b) $(6 + 4)^2$ (c) $2^3 + 3^2$ FLUENCY

2.4 · ORDER OF OPERATIONS

Brackets, fraction bars and the unary minus

Brackets are the writer's tool for changing the order. A fraction bar acts like brackets around the whole numerator and the whole denominator.



Evaluate the power first, then multiply, then add. Brackets could change this order.

WORKED EXAMPLE 8

Evaluate $(18 - 4 \times 2)^2 \div 5$ and $\frac{36-6}{2+4}$.

1. Inside the bracket: $4 \times 2 = 8$, so $18 - 8 = 10$. Then $10^2 \div 5 = 100 \div 5 = 20$.
2. Numerator: $36 - 6 = 30$. Denominator: $2 + 4 = 6$. Then $30 \div 6 = 5$.

20 and 5.

BOUNDARY NOTE

A minus sign in front of a power is not part of the base unless brackets include it: $-3^2 = -(3^2) = -9$, while $(-3)^2 = 9$.

GUIDED PRACTICE · COPY AND COMPLETE

Use these immediately after the matching worked examples. Check your reasoning, not only the final number.

G4 Boundary note: evaluate -3^2 and $(-3)^2$. Explain what the brackets include in the base. GUIDED

PRACTICE 2.4 · ORDER OF OPERATIONS

Show one operation per line when an expression has several steps. Multiplication and division share a rank; addition and subtraction share a rank.

2.4.4 Evaluate. (a) $24 \div 4 \times 2$ (b) $24 \div (4 \times 2)$ (c) $36 \div 6 + 3$ FLUENCY

2.4.5 Evaluate $5 \times 2^3 - 3^2$, showing each stage. FLUENCY

2.4 · ORDER OF OPERATIONS

Diagnose the error, then repair the calculation

When a calculation goes wrong, identify the first line where the convention was broken. Do not only change the final answer.

Incorrect

$12 + 6 \div 3 = 18 \div 3 = 6$

The student added before dividing.

Correct

$12 + 6 \div 3 = 12 + 2 = 14$

Division has higher priority than addition.

PRACTICE 2.4 · ORDER OF OPERATIONS

Show one operation per line when an expression has several steps. Multiplication and division share a rank; addition and subtraction share a rank.

2.4.6 Insert brackets to make each statement true. (a) 3 + 5 × 2 = 16 (b) 12 − 4 × 2 = 16 (c) 18 ÷ 3 + 6 = 2 REASONING

2.4.7 A student calculates 12 + 6 ÷ 3 as 18 ÷ 3 = 6. Identify the mistake, give the correct value, and explain which operation should be performed first. REASONING

2.4.8 Use the digits 2, 3 and 4 once each, with any operations and brackets, to make at least four different values. One expression must contain an index. CHALLENGE

COMMUNICATE

Write one operation per line. If a reader cannot see which rule you used, add a bracket or a short label rather than more words.

2.5 · SUBSTITUTION AND EXPRESSIONS WITH POWERS

Substitute first, then ask what is squared

A letter stands for a number. Substitution means replacing the letter with its value and evaluating using the order of operations. Keep brackets around the replacement so the structure stays visible.

$3x^2$ when x = 4

$3 \times 4^2 = 3 \times 16 = 48$

Only x is squared.

$(3x)^2$ when x = 4

$(3 \times 4)^2 = 12^2 = 144$

The whole product 3x is squared.

The brackets determine whether only x is squared or the whole product 3x is squared.

WORKED EXAMPLE 9

Evaluate $3x^2$ and $(3x)^2$ when $x = 4$.

1. $3x^2 = 3 \times 4^2 = 3 \times 16 = 48$.
2. $(3x)^2 = (3 \times 4)^2 = 12^2 = 144$.
3. The second expression squares the whole product, so it is three times the first.

48 and 144.

GUIDED PRACTICE · COPY AND COMPLETE

Use these immediately after the matching worked examples. Check your reasoning, not only the final number.

G3 For x = 5, compare $2x^2$ and $(2x)^2$. For x = 2, compare $x^2 + 3^2$ and $(x + 3)^2$. Explain what the brackets change. GUIDED

PRACTICE 2.5 · SUBSTITUTION AND POWERS

Replace the letter with its value in brackets, then evaluate. Ask what the exponent applies to.

2.5.1 Evaluate $2n^2$ for (a) n = 3 (b) n = 5 (c) n = 10. FLUENCY

2.5.2 Evaluate $(2n)^2$ for (a) n = 3 (b) n = 5 (c) n = 10. Compare each result with the matching result in question 2.5.1. FLUENCY

2.5.3 For a = 3 and b = 4, evaluate $a^2 + b^2$ and $(a + b)^2$. Are they equal? FLUENCY

2.5 · SUBSTITUTION AND EXPRESSIONS WITH POWERS

Formulas connect powers to measurement

Area and volume formulas use powers because they multiply equal dimensions. A model may use a power to describe growth or motion; always check what the variables and units mean.

WORKED EXAMPLE 10

Find the area of a square with $s = 7$ cm, the volume of a cube with $s = 4$ cm, and the model distance $d = 5t^2$ when $t = 3$.

1. $A = s^2 = 7^2 = 49$ cm^2.
2. $V = s^3 = 4^3 = 64$ cm^3.
3. $d = 5 \times 3^2 = 5 \times 9 = 45$ m. The index is evaluated before multiplying by 5.

49 cm^2, 64 cm^3, 45 m. The distance formula is a simplified model.

PRACTICE 2.5 · SUBSTITUTION AND POWERS

Replace the letter with its value in brackets, then evaluate. Ask what the exponent applies to.

2.5.4 A square has side length s = 12 cm. Use $A = s^2$ to find its area, with units. APPLICATION

2.5.5 Use $V = s^3$ to find the volume of a cube with edge (a) 3 cm (b) 5 cm. FLUENCY

2.5.6 In a simplified falling-object model, $d = 5t^2$, where d is in metres and t is in seconds. Find d when (a) t = 2 (b) t = 4. Explain why doubling t does not double d. REASONING

REASONING HABIT

Before calculating, say what the exponent applies to: a single variable, a product, or a sum.

2.5 · SUBSTITUTION AND EXPRESSIONS WITH POWERS

Test a transferred rule

A rule that works in one structure may fail in another. Test the claim $(a+b)^2 = a^2 + b^2$ before using it.

WORKED EXAMPLE 11

Test the claim with $a = 3, b = 4$, then with $a = 2, b = 5$.

1. $(3+4)^2 = 49$, but $3^2 + 4^2 = 25$.
2. $(2+5)^2 = 49$, but $2^2 + 5^2 = 29$.
3. One counterexample is enough to reject a universal claim.

The claim is false. Brackets change the base of the power.

PRACTICE 2.5 · SUBSTITUTION AND POWERS

Replace the letter with its value in brackets, then evaluate. Ask what the exponent applies to.

2.5.7 Insert one pair of brackets to make each statement true. (a) $2x^2 = 36$ when $x = 3$ (b) $a + b^2 = 49$ when $a = 2$ and $b = 5$. Explain how the brackets change the base. REASONING

2.5.8 A square has area 81 cm^2. Find its side length. If the side length is tripled, find the new area and explain how 3^2 appears in your answer. CHALLENGE

ATL · Transfer. In order for a student to decide whether a familiar rule applies, students must compare the structure of the new expression with the old one and verify the result with a substitution.

CHECKPOINT

Before the application

Answer quickly, then explain one answer to a partner. Use the answers at the back to check both the result and the reasoning.

CHECKPOINT · BEFORE THE APPLICATION

Answer quickly, then explain one answer to a partner.

C1 In 6^4, name the base and exponent, then evaluate the power. CHECK

C2 Write 5,020 in expanded form using powers of ten. CHECK

C3 Evaluate $\sqrt{144}$ and $\sqrt[3]{64}$. CHECK

C4 Evaluate $18 - 2 \times 3^2$. CHECK

Common slips

- Reading an exponent as a multiplier: $5^2 \neq 10$.
- Forgetting that $a^0 = 1$ only for nonzero a.
- Doing multiplication before division just because it appears first in a mnemonic.
- Squaring only part of a bracketed product or sum.

Explain how you know

- Count the factors in an expanded power.
- Name the place before writing a power-of-ten term.
- Reverse a root by squaring or cubing.
- Label the first operation you perform.

APPLICATION · CRITERION D

The school data store

Simulated context. A school media club in Cotonou is planning storage for video projects. Use the decimal units stated in the task: 1 TB = 1,000 GB. The figures are classroom planning data, not a real purchase order.

DRIVE	LABEL	CAPACITY IN GB
A	4 TB	4,000
B	4 TB	4,000
C	2 TB	2,000

APPLICATION · THE SCHOOL DATA STORE

Criterion D practice. This is a simulated planning task for a school media club in Cotonou. Use the stated decimal units: 1 TB = 1,000 GB. Show units and justify any rounding.

D1 The club plans to use three drives labelled 4 TB, 4 TB and 2 TB. Find the total capacity in TB and in GB, and write the GB total as a power of ten. APPLICATION

D2 The system reserves 15% of the total capacity. Each finished project needs 5 GB for a master file and 1 GB for a backup. How many complete projects can the club store after the reserve is removed, and how much space is left over? APPLICATION

D3 Explain why the answer in D2 must be rounded down rather than to the nearest whole number. REASONING

D4 Extension: some software reports storage in binary units. In this task, 1 TB = 10^{12} bytes and 1 TiB = 2^{40} bytes. Use a calculator to explain why 10 TB is about 9.09 TiB. Is any storage missing? Write one honest label for a poster. CHALLENGE

BEFORE YOU FINISH

Check that every answer has a unit, that complete projects are rounded down, and that any binary-unit comparison names the unit convention.

ASSESSMENT PRACTICE · CRITERIA A AND C

A power primer for the science lab

The science department wants a one-page reference card beside the microscopes. Write for students who have not yet studied this chapter. The card should be readable from about one metre away.

CRITERION A · KNOWING AND UNDERSTANDING
Use accurate definitions, notation and calculations.

CRITERION C · COMMUNICATING
Use mathematical language, diagrams, notation and a clear structure for the intended audience.

ASSESSMENT PRACTICE · A POWER PRIMER FOR THE SCIENCE LAB

Criteria A and C. Design a one-page reference card for students who have not yet studied this chapter. Use no more than 180 words of explanation, plus diagrams and one fully worked example.

S1 Define power, base and exponent using $5^3 = 125$ as your example. ASSESSMENT

S2 Explain 5^1 and 5^0 with a pattern that makes clear why $5^0 = 1$ rather than 0. ASSESSMENT

S3 Show $4^2 = 16$ as a square of 16 unit squares. Add a sketch or description explaining why 4^3 is called a cube. ASSESSMENT

S4 Work through $4 + 3 \times (8 - 2)^2 \div 6$, with one line per operation and a label naming the rule used at each step. ASSESSMENT

S5 Explain the difference between $3x^2$ and $(3x)^2$ when $x = 2$. Then ask a classmate to use your card to evaluate $2 + 5 \times 3^2$. Record one strength, one limitation and one revision. ASSESSMENT

PEER TEST
Ask a classmate to use your card to evaluate $2 + 5 \times 3^2$. If they cannot follow a step, revise the card rather than blaming the reader.

CHAPTER REVIEW

What you should now be able to do

Key ideas

- In a^n, the base is repeated and the exponent counts factors.
- $a^1 = a$; for $a \neq 0$, $a^0 = 1$.
- Powers of ten encode place value.
- Square roots and cube roots reverse squaring and cubing.
- Brackets, indices, then multiplication/division left to right, then addition/subtraction left to right.
- Substitution preserves structure: brackets change the base.

Vocabulary

power · base · exponent · index · square number · cube number · square root · cube root · expanded form · standard form · order of operations · substitution · formula

MIXED PRACTICE 2

Do not use a calculator unless a question asks for one. Show the order of operations for questions with more than one operation.

M1 Evaluate. (a) 3^4 (b) 2^7 (c) 10^6 (d) 6^3 (e) 11^2 (f) 4^4 FLUENCY

M2 Write as a single power. (a) $9 \times 9 \times 9$ (b) 12×12 (c) $10 \times 10 \times 10 \times 10 \times 10$ (d) $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$ FLUENCY

M3 Write 60,304 in expanded form using powers of ten. FLUENCY

M4 Write $8 \times 10^4 + 2 \times 10^2 + 5 \times 10^0$ in standard form. FLUENCY

M5 Evaluate. (a) $\sqrt{81}$ (b) $\sqrt{196}$ (c) $\sqrt[3]{27}$ (d) $\sqrt[3]{1{,}000}$ FLUENCY

M6 Calculate. (a) $6 + 2 \times 5$ (b) $(6 + 2) \times 5$ (c) $3^2 \times 4 - 10$ (d) $48 \div (4 + 4)$ (e) $5 + 12 \div 3$ (f) $(7 - 2)^2$ FLUENCY

MIXED PRACTICE 2

Fluency and reasoning

MIXED PRACTICE 2

Do not use a calculator unless a question asks for one. Show the order of operations for questions with more than one operation.

M7 Substitute $x = 3$ and $y = 2$. (a) $x^2 + y^2$ (b) $(x + y)^2$ (c) $2x^3$ (d) $(2x)^2$ FLUENCY

M8 Insert one pair of brackets to make each statement true. (a) $2 + 3 \times 4 = 20$ (b) $24 - 8 \div 4 = 4$ (c) $3 \times 2^2 = 36$ (d) $10 - 4^2 = 36$ REASONING

M9 Explain why $2^5 = 32$ but $5^2 = 25$. Which is larger, and why does the position of the exponent matter? REASONING

M10 A student writes $7^2 = 14$. Identify the error and give the correct value with a one-sentence explanation. REASONING

M11 Find the missing whole number. (a) $2^n = 64$ (b) $10^n = 100{,}000$ (c) $3^n = 81$ (d) $n^3 = 125$ REASONING

M12 Explain why $10^0 = 1$ using the pattern $10^3 = 1{,}000$, $10^2 = 100$, $10^1 = 10$. REASONING

EXPLAIN HOW YOU KNOW

Choose one answer and write a short justification using factors, place value, inverse operations or the order of operations.

MIXED PRACTICE 2 · REFLECTION

Challenge and reflect

MIXED PRACTICE 2

Do not use a calculator unless a question asks for one. Show the order of operations for questions with more than one operation.

M13 Evaluate $10^3 + 10^2 + 10^1 + 10^0$. Describe the pattern in the digits of the answer and explain why it occurs. REASONING

M14 A square rug has area 169 cm^2. Find its side length and perimeter. If the side length is doubled, find the new area and perimeter. Explain why the area changes by a factor of 2^2 while the perimeter changes by a factor of 2. APPLICATION

M15 A cube has edge 2 cm. Find its volume. Repeat for a cube with edge 4 cm. How many times larger is the second volume? Explain using 2^3. APPLICATION

M16 Using the numbers 4, 3 and 2 in that order, insert operations and brackets to make (a) 14 and (b) 20. Check each result with the order of operations. CHALLENGE

M17 A rumour spreads in rounds. One student tells 3 friends in round 1. In round 2, each of those 3 tells 3 new people, and the pattern continues. (a) How many new people hear it in round 4? (b) How many people in total have heard it after 4 rounds, including the original student? (c) Explain why the answer to (b) is not 3^5. CHALLENGE

M18 Explain why $\sqrt{(36 + 64)}$ is not $\sqrt{36} + \sqrt{64}$. Evaluate both expressions and state the rule that the example disproves. REASONING

Reflection

I can explain why an exponent is not a multiplier.

I can use powers of ten to justify expanded form.

I can reverse squares and cubes with roots.

I can show the order of operations one step at a time.

I can explain how brackets change the base of a power.

One rule I transferred correctly, and how I checked it: _____

One model limit I can explain: _____

CHAPTER 2 · ANSWERS AND REASONING

Answers · Ready check and Guided practice

READY CHECK

R1 **42 and 81.**
Six groups of seven make 42. Nine groups of nine make 81.

R2 **4,536.**
The terms occupy the thousands, hundreds, tens and ones places.

R3 **24.**
Multiply first: $3 \times 4 = 12$. Then add: $12 + 12 = 24$.

R4 **$3 \times 3 \times 3 = 27$.**
There are three factors of 3. First $3 \times 3 = 9$, then $9 \times 3 = 27$.

R5 **64 cm^2.**
Area of a square = side × side = $8 \times 8 = 64$ square centimetres.

R6 **$(2 + 4) \times 3 = 18$.**
The brackets make the addition happen first: $2 + 4 = 6$, then $6 \times 3 = 18$.

GUIDED PRACTICE · COPY AND COMPLETE

G1 **Base 8, exponent 3. $8 \times 8 \times 8 = 512$. If the exponent were 1, the value would be 8.**
The exponent counts factors of 8. It does not mean 8×3.

G2 **$\sqrt{169} = 13$; $\sqrt[3]{216} = 6$; side length = 14 cm.**
Roots reverse powers. Since $14^2=196$, a square with area 196 cm^2 has side 14 cm.

G3 **For x=5: $2x^2=50$ and $(2x)^2=100$. For x=2: $x^2+3^2=13$ and $(x+3)^2=25$. Brackets change the base of the power to include the whole product or sum.**
Without brackets, the exponent applies only to the quantity immediately before it. With brackets, it applies to everything inside the brackets.

G4 **$-3^2 = -9$, while $(-3)^2 = 9$. In the first expression the power is evaluated before the unary minus; in the second, the base is −3.**
-3^2 means $-(3^2)$. The brackets in $(-3)^2$ include the negative sign in the repeated factor: $(-3) \times (-3) = 9$.

CHAPTER 2 · ANSWERS AND REASONING

Answers · Practice 2.1

PRACTICE 2.1 · POWERS AND INDEX NOTATION

2.1.1 **(a) 4^3 (b) 7^2 (c) 10^5 (d) 2^7.**
The base is the repeated factor. The exponent counts how many copies of that factor are multiplied.

2.1.2 **(a) 8 (b) 9 (c) 125 (d) 1,000 (e) 64 (f) 64.**
Expand first if needed: 2×2×2=8; 3×3=9; 5×5×5=125; 10×10×10=1,000; six 2s make 64; 4×4×4=64.

2.1.3 **(a) 4 (b) 6 (c) 3 (d) 0.**
3^4=81, 2^6=64 and 10^3=1,000. For the nonzero base 7, the exponent 0 gives 1.

2.1.4 **4^2 = 16, but 4 × 2 = 8. The exponent counts equal factors; it is not a multiplier.**
4^2 means 4 × 4. The expression 4 × 2 means two groups of 4, or 4 + 4.

2.1.5 **9^2 = 81 cm².**
A square's area is side × side. The unit is square centimetres because two lengths in centimetres are multiplied.

2.1.6 **$2^4 = 4^2 = 16$, but $2^3 = 8$ and $3^2 = 9$. The claim is false; one counterexample is enough.**
The first pair happens to match. The second pair does not, so a universal claim about swapping base and exponent cannot be true.

2.1.7 **Round 4: 3^4 = 81 new people. Rounds 1–4: 3 + 9 + 27 + 81 = 120 new people.**
Each round multiplies the number of new people by 3. The total asked for excludes the original person.

2.1.8 **2^{10} is larger. Exactly, $2^{10} = 1{,}024$ and $10^3 = 1{,}000$.**
One check is $2^{10} = 2^7 \times 2^3 = 128 \times 8$. Since 128 > 125, multiplying both by 8 gives 128 × 8 > 125 × 8 = 1,000. The exact value is 1,024.

CHAPTER 2 · ANSWERS AND REASONING

Answers · Practice 2.2

PRACTICE 2.2 · POWERS OF TEN AND EXPANDED FORM

2.2.1 **(a) 100 (b) 10,000 (c) 1,000,000 (d) 1.**

For positive whole-number exponents, 10^n is 1 followed by n zeros. The pattern of dividing by 10 gives $10^0 = 1$.

2.2.2 **$8 \times 10^3 + 3 \times 10^2 + 0 \times 10^1 + 4 \times 10^0$.**

The digits occupy thousands, hundreds, tens and ones. The zero tens term may be shown to make the place structure explicit.

2.2.3 **4,706.**

The terms are 4,000, 700, 0 and 6. The zero keeps the tens place empty.

2.2.4 **$9 \times 10^4 + 0 \times 10^3 + 0 \times 10^2 + 5 \times 10^1 + 0 \times 10^0$; the 5 is worth 50.**

The 5 is in the tens place. Zeros hold the thousands, hundreds and ones places.

2.2.5 **20,050. Zeros hold the thousands, hundreds and ones places.**

$2 \times 10^4 = 20{,}000$ and $5 \times 10^1 = 50$. Adding them gives 20,050; the missing places must not be collapsed.

2.2.6 **$1 \times 10^4 + 2 \times 10^3 + 4 \times 10^2 + 8 \times 10^1 + 0 \times 10^0$. The 1 contributes 10,000, the largest share.**

Compare the values of the nonzero terms: 10,000, 2,000, 400 and 80. The ten-thousands term is largest.

2.2.7 **3,000 is 10 times 300.**

The digit 3 has moved one place to the left. Each place is ten times the place to its right.

2.2.8 **Sample: $12{,}690 = 1 \times 10^4 + 2 \times 10^3 + 6 \times 10^2 + 9 \times 10^1 + 0 \times 10^0$. The digits sum to 18 and there is exactly one zero term. Many answers are possible.**

Choose four nonzero digits that sum to 18, then place one zero in a non-leading position. The leading digit cannot be zero in a five-digit number.

CHAPTER 2 · ANSWERS AND REASONING

Answers · Practices 2.3 and 2.4

PRACTICE 2.3 · SQUARES, CUBES AND ROOTS

2.3.1 **1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.**
These are 1^2 through 12^2. Each is a whole number multiplied by itself.

2.3.2 **(a) 169 (b) 225 (c) 216 (d) 1,000.**
13×13=169; 15×15=225; 6×6×6=216; 10×10×10=1,000.

2.3.3 **(a) 7 (b) 10 (c) 13 (d) 2 (e) 3 (f) 5.**
Reverse the powers: 7^2=49, 10^2=100, 13^2=169, 2^3=8, 3^3=27 and 5^3=125.

2.3.4 **12 tiles.**
If each side has n tiles, the square contains n^2 tiles. Since $12^2 = 144$, n = 12.

2.3.5 **(a) 8 (b) 6 (c) 15.**
8^2=64, 6^3=216 and 15^2=225. The question asks for the positive value.

2.3.6 **$\sqrt{81} + \sqrt{16} = 9 + 4 = 13$. $\sqrt{(81 + 16)} = \sqrt{97}$, which is not 13 because $13^2 = 169$. They are not equal.**
In the first expression the roots are taken separately and then added. In the second, the sum is formed first and the root applies to the whole sum.

2.3.7 **Between 7 and 8, closer to 7.**
7^2=49 and 8^2=64, so $7 < \sqrt{50} < 8$. Also 7.5^2=56.25, and $50 < 56.25$, so $\sqrt{50} < 7.5$ and is closer to 7.

2.3.8 **Edge length 7 cm. Doubled edge 14 cm gives $14^3 = 2{,}744$ cm³, which is 8×343. The factor is $2^3 = 8$.**
7^3=343. Doubling each of the three dimensions multiplies volume by 2 × 2 × 2.

PRACTICE 2.4 · ORDER OF OPERATIONS

2.4.1 **18.**
Multiply first: 2 × 5 = 10. Then add: 8 + 10 = 18.

2.4.2 **8.**
Multiply first: 3 × 4 = 12. Then subtract: 20 − 12 = 8.

2.4.3 **(a) 22 (b) 100 (c) 17.**
(a) 4^2=16, then 6+16=22. (b) Brackets first: 10^2=100. (c) 8+9=17.

2.4.4 **(a) 12 (b) 3 (c) 9.**
(a) Division and multiplication have equal priority, so work left to right: 24÷4=6, then 6×2=12. (b) Brackets first: 4×2=8, then 24÷8=3. (c) 36÷6=6, then 6+3=9.

2.4.5 **$5 \times 8 - 9 = 40 - 9 = 31$.**
Evaluate the powers first: 2^3=8 and 3^2=9. Then multiply, then subtract.

2.4.6 **(a) $(3 + 5) \times 2 = 16$ (b) $(12 - 4) \times 2 = 16$ (c) $18 \div (3 + 6) = 2$.**
The brackets change which operation is completed first. Check each result after inserting them.

2.4.7 **The student added before dividing. Correct: $6 \div 3 = 2$, then $12 + 2 = 14$.**
Division has higher priority than addition unless brackets say otherwise.

2.4.8 **Samples: $2 + 3 \times 4 = 14$; $(2 + 3) \times 4 = 20$; $2^3 + 4 = 12$; $4^2 - 3 = 13$. Other valid expressions are possible.**
Each sample uses 2, 3 and 4 exactly once. The order of operations and brackets determine the value.

CHAPTER 2 · ANSWERS AND REASONING

Answers · Practice 2.5 and Checkpoint

PRACTICE 2.5 · SUBSTITUTION AND POWERS

2.5.1 **(a) 18 (b) 50 (c) 200.**
Square n first, then multiply by 2: 2×9=18; 2×25=50; 2×100=200.

2.5.2 **(a) 36 (b) 100 (c) 400. Each is double the matching result in 2.5.1.**
The brackets include the factor 2 in the base: $(2n)^2 = 4n^2$, while $2n^2$ is only $2n^2$. Therefore $(2n)^2$ is twice $2n^2$.

2.5.3 **$a^2 + b^2 = 9 + 16 = 25$. $(a + b)^2 = 7^2 = 49$. They are not equal.**
The first expression squares a and b separately. The second squares the whole sum.

2.5.4 **$A = 12^2 = 144$ cm².**
Substitute 12 for s. Multiplying centimetres by centimetres gives square centimetres.

2.5.5 **(a) 27 cm^3 (b) 125 cm^3.**
3^3=27 and 5^3=125. Multiplying three lengths in centimetres gives cubic centimetres.

2.5.6 **(a) 20 m (b) 80 m. Doubling t multiplies t^2 by 4, so d quadruples in this model.**
$5×2^2$=20 and $5×4^2$=80. The model is simplified; it is not a precise measurement of a real fall.

2.5.7 **(a) $(2x)^2 = 36$ (b) $(a + b)^2 = 49$. The brackets put the whole product or sum in the base of the power.**
When x=3, $(2×3)^2=6^2$=36. When a=2 and b=5, $(2+5)^2=7^2$=49.

2.5.8 **Side length 9 cm. Tripled side 27 cm gives new area $27^2 = 729$ cm² $= 9 × 81$. The area is multiplied by $3^2 = 9$.**
Area depends on the square of the side length. Tripling the side multiplies area by 3 × 3.

CHECKPOINT · BEFORE THE APPLICATION

C1 **Base 6, exponent 4; $6^4 = 1{,}296$.**
$6^4 = 6×6×6×6 = 36×36 = 1{,}296$.

C2 **$5 × 10^3 + 0 × 10^2 + 2 × 10^1 + 0 × 10^0$.**
The zeros hold the hundreds and ones places.

C3 **12 and 4.**
12^2=144 and 4^3=64.

C4 **0.**
3^2=9, then 2×9=18, then 18−18=0.

CHAPTER 2 · ANSWERS AND REASONING

Answers · Exploration · The doubling sheet

EXPLORATION · THE DOUBLING SHEET

B1 **Physical results vary. Ideal model: layers 1, 2, 4, 8, 16, 32; thicknesses 0.1, 0.2, 0.4, 0.8, 1.6, 3.2 mm for n = 0, 1, 2, 3, 4, 5.**

Start with one layer. Each fold doubles the layer count. Thickness = 0.1 × layers. The physical trial and the ideal model are different kinds of evidence.

B2 **The thickness doubles with each fold. $T = 0.1 \times 2^n$ mm.**

After n folds there are 2^n layers. At n=0, 2^0=1, so the rule gives the original thickness 0.1 mm.

B3 **$T = 0.1 \times 2^{10} = 102.4$ mm. It is 1,024 times thicker, because $2^{10} = 1{,}024$.**

The starting thickness is 0.1 mm. Dividing 102.4 by 0.1 gives 1,024, the layer-count factor.

B4 **14 folds. At 13 folds, T = 819.2 mm; at 14 folds, T = 1,638.4 mm. Since 1 m = 1,000 mm, 14 is the first fold count above 1 m.**

Use $T = 0.1 \times 2^n$. Check n=13 and n=14 rather than guessing from the size of the exponent.

B5 **No. The table is an ideal model, not proof of physical feasibility. Assumptions include perfect folds, constant thickness and enough sheet area to keep folding. Improve the estimate by measuring a stack of many sheets and dividing by the number of sheets.**

Real paper resists folding because thickness grows, creases form and the available area shrinks. A mathematical prediction can be valid inside a model while the model stops matching reality.

B6 **Tripling model: $0.1 \times 3^5 = 24.3$ mm. Doubling model: $0.1 \times 2^5 = 3.2$ mm. The tripling model grows faster because each step multiplies by 3 rather than 2.**

Compare the growth factors, not only the starting value. Both start at 0.1 mm when n=0.

CHAPTER 2 · ANSWERS AND REASONING

Answers · Application and Assessment practice

APPLICATION · THE SCHOOL DATA STORE

D1 **10 TB = 10,000 GB = 10^4 GB.**
4+4+2=10 TB. With 1 TB=1,000 GB, multiply by 1,000: 10×1,000=10,000 GB.

D2 **1,416 complete projects, with 4 GB left over.**
15% means 15 out of 100, so the reserve is 10,000 × 15 ÷ 100 = 1,500 GB. Available space is 8,500 GB. Each project needs 6 GB. 8,500 ÷ 6 = 1,416 remainder 4.

D3 **A 1,417th project would need 1,417 × 6 = 8,502 GB, but only 8,500 GB is available. Complete projects require rounding down.**
Nearest-whole rounding can create a capacity that does not exist. The context requires a feasible whole number of complete projects.

D4 **10 TB = 10^{13} bytes. $10^{13} \div 2^{40} \approx 9.09$ TiB. No storage is missing; the units differ. Sample label: '10 TB decimal capacity (about 9.09 TiB in binary units).'**
A tebibyte is a larger unit than a terabyte, so the same number of bytes gives a smaller count in TiB. Honesty means naming the unit, not choosing the larger number.

ASSESSMENT PRACTICE · A POWER PRIMER FOR THE SCIENCE LAB

S1 **Sample: 'A power records repeated multiplication. In 5^3, 5 is the base and 3 is the exponent. It means 5 × 5 × 5 = 125.'**
The definition names the repeated factor and the number of factors, and the example matches the notation.

S2 **Sample: '5^3=125, 5^2=25, 5^1=5. Each step divides by 5, so the next step gives 5^0=1.'**
The pattern continues the division-by-the-base rule. The zero-exponent rule applies because the base 5 is nonzero.

S3 **Sample: draw a 4 by 4 grid and label it 4^2=16. For 4^3, draw or describe 4 layers of 4 by 4 unit cubes: 4 × 4 × 4 = 64 unit cubes.**
The square model shows two equal dimensions; the cube model shows three equal dimensions.

S4 **Brackets: 8−2=6. Index: 6^2=36. Multiply: 3×36=108. Divide: 108÷6=18. Add: 4+18=22.**
Brackets first, then indices, then multiplication/division from left to right, then addition/subtraction. The final value is 22.

S5 **When x=2, $3x^2$=12 and $(3x)^2$=36. The test expression is 2 + 5 × 9 = 47. Sample evaluation: strength—one step per line; limitation—symbols may be too small; revision—enlarge the 'indices before multiplication' label.**
The card should make the base of each power visible and support a reader who is learning the convention for the first time.

CHAPTER 2 · ANSWERS AND REASONING

Answers · Mixed practice 2 (M1–M9)

MIXED PRACTICE 2

M1 **(a) 81 (b) 128 (c) 1,000,000 (d) 216 (e) 121 (f) 256.**
Expand or use known squares/cubes: $3^4=81$; $2^7=128$; $10^6=1{,}000{,}000$; $6^3=216$; $11^2=121$; $4^4=256$.

M2 **(a) 9^3 (b) 12^2 (c) 10^5 (d) 5^7.**
Count the repeated factors. The base stays the same.

M3 **$6 \times 10^4 + 0 \times 10^3 + 3 \times 10^2 + 0 \times 10^1 + 4 \times 10^0$.**
The zero terms hold the thousands and tens places.

M4 **80,205.**
The terms are 80,000, 200 and 5. Zeros hold the thousands and tens places.

M5 **(a) 9 (b) 14 (c) 3 (d) 10.**
$9^2=81$, $14^2=196$, $3^3=27$ and $10^3=1{,}000$.

M6 **(a) 16 (b) 40 (c) 26 (d) 6 (e) 9 (f) 25.**
Apply brackets, indices, multiplication/division left to right, then addition/subtraction left to right.

M7 **(a) 13 (b) 25 (c) 54 (d) 36.**
(a) 9+4=13. (b) $5^2=25$. (c) 2×27=54. (d) $6^2=36$.

M8 **(a) $(2 + 3) \times 4 = 20$ (b) $(24 - 8) \div 4 = 4$ (c) $(3 \times 2)^2 = 36$ (d) $(10 - 4)^2 = 36$.**
The brackets change the grouping and, in (c) and (d), the base of the power.

M9 **2^5 uses five factors of 2, giving 32. 5^2 uses two factors of 5, giving 25. Therefore 2^5 is larger. The base and exponent play different roles.**
Swapping base and exponent changes both the repeated factor and the number of factors. The correct comparison is $32 > 25$.

CHAPTER 2 · ANSWERS AND REASONING

Answers · Mixed practice 2 (M10–M18)

MIXED PRACTICE 2

M10 **$7^2 = 49$, not 14. The student multiplied 7 by 2 instead of using two factors of 7.**
The exponent counts factors: $7^2 = 7 \times 7$.

M11 **(a) 6 (b) 5 (c) 4 (d) 5.**
2^6=64; 10^5=100,000; 3^4=81; 5^3=125.

M12 **Each step down in exponent divides by 10: 1,000 ÷ 100 ÷ 10 ÷ 1. Therefore $10^0 = 1$.**
The zero exponent continues the place-value pattern. It is not the same as multiplying by zero.

M13 **1,111. Each power of ten contributes a 1 in a different place: thousands, hundreds, tens and ones.**
1,000 + 100 + 10 + 1 = 1,111.

M14 **Side 13 cm, perimeter 52 cm. Doubled side 26 cm gives area 676 cm^2 and perimeter 104 cm. Area is multiplied by $4 = 2^2$; perimeter is multiplied by 2.**
√169=13. Perimeter is 4s, so it scales linearly. Area is s^2, so it scales by the square of the scale factor.

M15 **8 cm^3 and 64 cm^3. The second is 8 times larger because doubling each of three dimensions multiplies volume by $2^3 = 8$.**
2^3=8 and 4^3=64; 64÷8=8.

M16 **Samples: (a) $4 \times 3 + 2 = 14$ (b) $4 \times (3 + 2) = 20$.**
In (a), multiply before adding. In (b), brackets make the sum 5 before multiplying by 4.

M17 **(a) 81 new people. (b) 1 + 3 + 9 + 27 + 81 = 121 people. (c) $3^5 = 243$ would describe the number of new people in round 5, not the total after four rounds.**
New people by round are 3^1, 3^2, 3^3, 3^4. The total includes the original person and stops after round 4.

M18 **$\sqrt{(36 + 64)} = \sqrt{100} = 10$, while $\sqrt{36} + \sqrt{64} = 6 + 8 = 14$. The example disproves the claim that a square root can be distributed over a sum.**
The root applies to the whole value under the radical. Adding first and then taking the root is different from taking roots first and then adding.